ASSIGNMENT 2

```
mysql> create database A2;
Query OK, 1 row affected (0.01 sec)
mysql> use A2;
Database changed
mysql> create table employees(employee_id int primary key,first_name varchar(30),last_name varchar(30),
email varchar(50),phone_number varchar(20),hire_date date,job_id varchar(20),salary decimal(10,2),
department_id int);
Query OK, 0 rows affected (0.02 sec)
mysql> INSERT INTO employees
 -> (employee_id, first_name, last_name, email, phone_number, hire_date, job_id, salary, department_id)
 -> VALUES
 -> (101, 'Rahul', 'Sharma', 'rahul.sharma@gmail.com', '9876543201', '2022-01-15', 'IT_PROG', 55000, 60),
 -> (102, 'Priya', 'Mehta', 'priya.mehta@gmail.com', '9876543202', '2021-03-20', 'IT_PROG', 62000, 60),
 -> (103, 'Amit', 'Patel', 'amit.patel@gmail.com', '9876543203', '2023-05-10', 'SA_REP', 48000, 80),
 -> (104, 'Neha', 'Verma', 'neha.verma@gmail.com', '9876543204', '2020-07-18', 'HR_REP', 52000, 40),
 -> (105, 'Rohan', 'Chauhan', 'rohan.chauhan@gmail.com', '9876543205', '2022-11-05', 'IT_PROG', 65000, 60),
 -> (106, 'Sneha', 'Parihar', 'sneha.parihar@gmail.com', '9876543206', '2024-02-12', 'IT_PROG', 58000, 60),
 -> (107, 'Karan', 'Joshi', 'karan.joshi@gmail.com', '9876543207', '2021-09-25', 'SA_REP', 47000, 80),
 -> (108, 'Pooja', 'Bansal', 'pooja.bansal@gmail.com', '9876543208', '2023-12-01', 'HR_REP', 50000, 40),
 -> (109, 'Vikas', 'Singh', 'vikas.singh@gmail.com', '9876543209', '2019-04-14', 'IT_PROG', 72000, 60),
 -> (110, 'Anjali', 'Kapoor', 'anjali.kapoor@gmail.com', '9876543210', '2022-06-22', 'SA_REP', 49000, 80),
 -> (111, 'Arjun', 'Shetty', 'arjun.shetty@gmail.com', '9876543211', '2020-10-10', 'IT_PROG', 68000, 60),
 -> (112, 'Kavita', 'Shukla', 'kavita.shukla@gmail.com', '9876543212', '2021-08-17', 'HR_REP', 54000, 40),
 -> (113, 'Nitin', 'Kher', 'nitin.kher@gmail.com', '9876543213', '2023-01-30', 'SA_REP', 46000, 80),
 -> (114, 'Riya', 'Chopra', 'riya.chopra@gmail.com', '9876543214', '2024-03-15', 'IT_PROG', 61000, 60),
 -> (115, 'Mohit', 'Bhardwaj', 'mohit.bhardwaj@gmail.com', '9876543215', '2020-12-20', 'SA_REP', 51000, 80),
 -> (116, 'Simran', 'Kaur', 'simran.kaur@gmail.com', '9876543216', '2022-04-11', 'HR_REP', 56000, 40),
 -> (117, 'Varun', 'Bedi', 'varun.bedi@gmail.com', '9876543217', '2023-07-19', 'IT_PROG', 59000, 60),
 -> (118, 'Nisha', 'Desai', 'nisha.desai@gmail.com', '9876543218', '2021-02-28', 'SA_REP', 53000, 80),
 -> (119, 'Saurabh', 'Jain', 'saurabh.jain@gmail.com', '9876543219', '2019-09-09', 'IT_PROG', 75000, 60),
 -> (120, 'Divya', 'Malhotra', 'divya.malhotra@gmail.com', '9876543220', '2024-05-06', 'HR_REP', 57000, 40);
Query OK, 20 rows affected (0.01 sec)
Records: 20  Duplicates: 0  Warnings: 0
```

mysql> select *from employees;

employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	department_id
101	Rahul	Sharma	rahul.sharma@gmail.com	9876543201	2022-01-15	IT_PROG	55000.00	60
102	Priya	Mehta	priya.mehta@gmail.com	9876543202	2021-03-20	IT_PROG	62000.00	60
103	Amit	Patel	amit.patel@gmail.com	9876543203	2023-05-10	SA_REP	48000.00	80
104	Neha	Verma	neha.verma@gmail.com	9876543204	2020-07-18	HR_REP	52000.00	40
105	Rohan	Chauhan	rohan.chauhan@gmail.com	9876543205	2022-11-05	IT_PROG	65000.00	60
106	Sneha	Parihar	sneha.parihar@gmail.com	9876543206	2024-02-12	IT_PROG	58000.00	60
107	Karan	Joshi	karan.joshi@gmail.com	9876543207	2021-09-25	SA_REP	47000.00	80
108	Pooja	Bansal	pooja.bansal@gmail.com	9876543208	2023-12-01	HR_REP	50000.00	40
109	Vikas	Singh	vikas.singh@gmail.com	9876543209	2019-04-14	IT_PROG	72000.00	60
110	Anjali	Kapoor	anjali.kapoor@gmail.com	9876543210	2022-06-22	SA_REP	49000.00	80
111	Arjun	Shetty	arjun.shetty@gmail.com	9876543211	2020-10-10	IT_PROG	68000.00	60
112	Kavita	Shukla	kavita.shukla@gmail.com	9876543212	2021-08-17	HR_REP	54000.00	40
113	Nitin	Kher	nitin.kher@gmail.com	9876543213	2023-01-30	SA_REP	46000.00	80
114	Riya	Chopra	riya.chopra@gmail.com	9876543214	2024-03-15	IT_PROG	61000.00	60
115	Mohit	Bhardwaj	mohit.bhardwaj@gmail.com	9876543215	2020-12-20	SA_REP	51000.00	80
116	Simran	Kaur	simran.kaur@gmail.com	9876543216	2022-04-11	HR_REP	56000.00	40
117	Varun	Bedi	varun.bedi@gmail.com	9876543217	2023-07-19	IT_PROG	59000.00	60
118	Nisha	Desai	nisha.desai@gmail.com	9876543218	2021-02-28	SA_REP	53000.00	80
119	Saurabh	Jain	saurabh.jain@gmail.com	9876543219	2019-09-09	IT_PROG	75000.00	60
120	Divya	Malhotra	divya.malhotra@gmail.com	9876543220	2024-05-06	HR_REP	57000.00	40

ASSIGNMENT 2

QUES1:

mysql> select first_name,last_name from employees;

first_name	last_name
Rahul	Sharma
Priya	Mehta
Amit	Patel
Neha	Verma
Rohan	Chauhan
Sneha	Parihar
Karan	Joshi
Pooja	Bansal
Vikas	Singh
Anjali	Kapoor
Arjun	Shetty
Kavita	Shukla
Nitin	Kher
Riya	Chopra
Mohit	Bhardwaj
Simran	Kaur
Varun	Bedi
Nisha	Desai
Saurabh	Jain
Divya	Malhotra

20 rows in set (0.00 sec)

QUES2:

mysql> select distinct department_id from employees;

department_id
60
80
40

3 rows in set (0.00 sec)

ASSIGNMENT 2

QUES 3:

mysql> select * from employees order by first_name desc;

employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	department_id
109	Vikas	Singh	vikas.singh@gmail.com	9876543209	2019-04-14	IT_PROG	72000.00	60
117	Varun	Bedi	varun.bedi@gmail.com	9876543217	2023-07-19	IT_PROG	59000.00	60
106	Sneha	Parihar	sneha.parihar@gmail.com	9876543206	2024-02-12	IT_PROG	58000.00	60
116	Simran	Kaur	simran.kaur@gmail.com	9876543216	2022-04-11	HR_REP	56000.00	40
119	Saurabh	Jain	saurabh.jain@gmail.com	9876543219	2019-09-09	IT_PROG	75000.00	60
105	Rohan	Chauhan	rohan.chauhan@gmail.com	9876543205	2022-11-05	IT_PROG	65000.00	60
114	Riya	Chopra	riya.chopra@gmail.com	9876543214	2024-03-15	IT_PROG	61000.00	60
101	Rahul	Sharma	rahul.sharma@gmail.com	9876543201	2022-01-15	IT_PROG	55000.00	60
102	Priya	Mehta	priya.mehta@gmail.com	9876543202	2021-03-20	IT_PROG	62000.00	60
108	Pooja	Bansal	pooja.bansal@gmail.com	9876543208	2023-12-01	HR_REP	50000.00	40
113	Nitin	Kher	nitin.kher@gmail.com	9876543213	2023-01-30	SA_REP	46000.00	80
118	Nisha	Desai	nisha.desai@gmail.com	9876543218	2021-02-28	SA_REP	53000.00	80
104	Neha	Verma	neha.verma@gmail.com	9876543204	2020-07-18	HR_REP	52000.00	40
115	Mohit	Bhardwaj	mohit.bhardwaj@gmail.com	9876543215	2020-12-20	SA_REP	51000.00	80
112	Kavita	Shukla	kavita.shukla@gmail.com	9876543212	2021-08-17	HR_REP	54000.00	40
107	Karan	Joshi	karan.joshi@gmail.com	9876543207	2021-09-25	SA_REP	47000.00	80
120	Divya	Malhotra	divya.malhotra@gmail.com	9876543220	2024-05-06	HR_REP	57000.00	40
111	Arjun	Shetty	arjun.shetty@gmail.com	9876543211	2020-10-10	IT_PROG	68000.00	60
110	Anjali	Kapoor	anjali.kapoor@gmail.com	9876543210	2022-06-22	SA_REP	49000.00	80
103	Amit	Patel	amit.patel@gmail.com	9876543203	2023-05-10	SA_REP	48000.00	80

20 rows in set (0.00 sec)

QUES 4:

mysql> select first_name,last_name,salary,salary*0.15 as pf from employees;

first_name	last_name	salary	pf
Rahul	Sharma	55000.00	8250.0000
Priya	Mehta	62000.00	9300.0000
Amit	Patel	48000.00	7200.0000
Neha	Verma	52000.00	7800.0000
Rohan	Chauhan	65000.00	9750.0000
Sneha	Parihar	58000.00	8700.0000
Karan	Joshi	47000.00	7050.0000
Pooja	Bansal	50000.00	7500.0000
Vikas	Singh	72000.00	10800.0000
Anjali	Kapoor	49000.00	7350.0000
Arjun	Shetty	68000.00	10200.0000
Kavita	Shukla	54000.00	8100.0000
Nitin	Kher	46000.00	6900.0000
Riya	Chopra	61000.00	9150.0000
Mohit	Bhardwaj	51000.00	7650.0000
Simran	Kaur	56000.00	8400.0000
Varun	Bedi	59000.00	8850.0000
Nisha	Desai	53000.00	7950.0000
Saurabh	Jain	75000.00	11250.0000
Divya	Malhotra	57000.00	8550.0000

20 rows in set (0.00 sec)

QUES 5:

mysql> select employee_id,first_name,last_name,salary from employees order by salary asc;

employee_id	first_name	last_name	salary
113	Nitin	Kher	46000.00
107	Karan	Joshi	47000.00
103	Amit	Patel	48000.00
110	Anjali	Kapoor	49000.00
108	Pooja	Bansal	50000.00
115	Mohit	Bhardwaj	51000.00
104	Neha	Verma	52000.00
118	Nisha	Desai	53000.00
112	Kavita	Shukla	54000.00
101	Rahul	Sharma	55000.00
116	Simran	Kaur	56000.00
120	Divya	Malhotra	57000.00
106	Sneha	Parihar	58000.00
117	Varun	Bedi	59000.00
114	Riya	Chopra	61000.00
102	Priya	Mehta	62000.00
105	Rohan	Chauhan	65000.00
111	Arjun	Shetty	68000.00
109	Vikas	Singh	72000.00
119	Saurabh	Jain	75000.00

QUES 6:

mysql> select sum(salary) as 'total salary'from employees;

total salary
1138000.00

1 row in set (0.00 sec)

QUES 7:

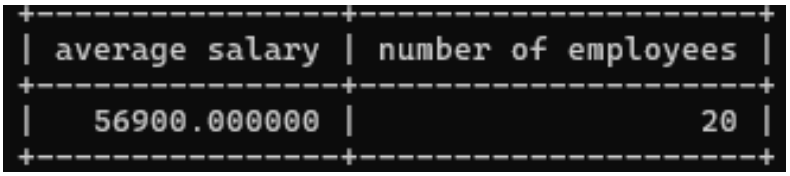
mysql> select max(salary) as 'maximum salary',min(salary) as 'minimum salary' from employees;

maximum salary	minimum salary
75000.00	46000.00

1 row in set (0.00 sec)

QUES 8:

mysql> select avg(salary) as'average salary',count(*) as'number of employees'from employees;

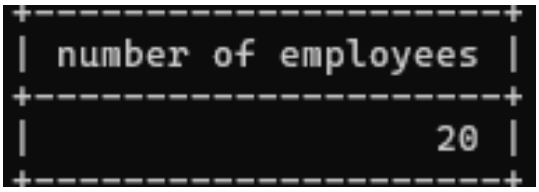


average salary	number of employees
56900.000000	20

1 row in set (0.00 sec)

QUES 9:

mysql> select count(*) as 'number of employees' from employees;

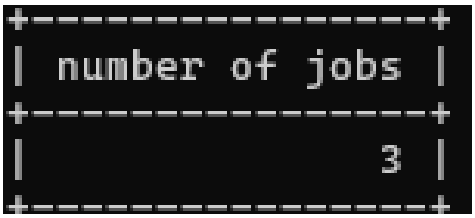


number of employees
20

1 row in set (0.01 sec)

QUES 10:

mysql> select count(distinct job_id) as 'number of jobs' from employees;



number of jobs
3

1 row in set (0.00 sec)

QUES 11:

mysql> select upper(first_name) as 'first_name' from employees;

first_name
RAHUL
PRIYA
AMIT
NEHA
ROHAN
SNEHA
KARAN
POOJA
VIKAS
ANJALI
ARJUN
KAVITA
NITIN
RIYA
MOHIT
SIMRAN
VARUN
NISHA
SAURABH
DIVYA

20 rows in set (0.00 sec)

QUES 12:

mysql> select substring(first_name,1,3) as 'first 3 character' from employees;

first 3 character
Rah
Pri
Ami
Neh
Roh
Sne
Kar
Poo
Vik
Anj
Arj
Kav
Nit
Riy
Moh
Sim
Var
Nis
Sau
Div

20 rows in set (0.00 sec)

QUES 13:

mysql> select 171*214+625 as result;

result
37219

QUES 14:

mysql> select concat(first_name,'',last_name) as 'employee name' from employees;



```
+---------------+
| employee name |
+---------------+
| RahulSharma   |
| PriyaMehta    |
| AmitPatel     |
| NehaVerma     |
| RohanChauhan  |
| SnehaParihar  |
| KaranJoshi    |
| PoojaBansal   |
| VikasSingh    |
| AnjaliKapoor  |
| ArjunShetty   |
| KavitaShukla  |
| NitinKher     |
| RiyaChopra    |
| MohitBhardwaj |
| SimranKaur    |
| VarunBedi     |
| NishaDesai    |
| SaurabhJain   |
| DivyaMalhotra |
+---------------+
```

20 rows in set (0.01 sec)

QUES 15:

mysql> select trim(first_name) as 'first_name' from employees;



```
+------------+
| first_name |
+------------+
| Rahul      |
| Priya      |
| Amit       |
| Neha       |
| Rohan      |
| Sneha      |
| Karan      |
| Pooja      |
| Vikas      |
| Anjali     |
| Arjun      |
| Kavita     |
| Nitin      |
| Riya       |
| Mohit      |
| Simran     |
| Varun      |
| Nisha      |
| Saurabh    |
| Divya      |
+------------+
```

20 rows in set (0.00 sec)

QUES 16:

mysql> select first_name,last_name,length(concat(first_name, ' ',last_name)) as 'name length' from employees;

first_name	last_name	name length
Rahul	Sharma	13
Priya	Mehta	12
Amit	Patel	11
Neha	Verma	11
Rohan	Chauhan	14
Sneha	Parihar	14
Karan	Joshi	12
Pooja	Bansal	13
Vikas	Singh	12
Anjali	Kapoor	14
Arjun	Shetty	13
Kavita	Shukla	14
Nitin	Kher	11
Riya	Chopra	12
Mohit	Bhardwaj	15
Simran	Kaur	12
Varun	Bedi	11
Nisha	Desai	12
Saurabh	Jain	13
Divya	Malhotra	15

20 rows in set (0.00 sec)

QUES 17:

mysql> select first_name from employees where first_name regexp '[0-9]';

Empty set (0.02 sec)

QUES 18:

mysql> select * from employees limit 10;

employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	department_id
101	Rahul	Sharma	rahul.sharma@gmail.com	9876543201	2022-01-15	IT_PROG	55000.00	60
102	Priya	Mehta	priya.mehta@gmail.com	9876543202	2021-03-20	IT_PROG	62000.00	60
103	Amit	Patel	amit.patel@gmail.com	9876543203	2023-05-10	SA_REP	48000.00	80
104	Neha	Verma	neha.verma@gmail.com	9876543204	2020-07-18	HR_REP	52000.00	40
105	Rohan	Chauhan	rohan.chauhan@gmail.com	9876543205	2022-11-05	IT_PROG	65000.00	60
106	Sneha	Parihar	sneha.parihar@gmail.com	9876543206	2024-02-12	IT_PROG	58000.00	60
107	Karan	Joshi	karan.joshi@gmail.com	9876543207	2021-09-25	SA_REP	47000.00	80
108	Pooja	Bansal	pooja.bansal@gmail.com	9876543208	2023-12-01	HR_REP	50000.00	40
109	Vikas	Singh	vikas.singh@gmail.com	9876543209	2019-04-14	IT_PROG	72000.00	60
110	Anjali	Kapoor	anjali.kapoor@gmail.com	9876543210	2022-06-22	SA_REP	49000.00	80

10 rows in set (0.00 sec)

QUES 19:

mysql> select first_name, last_name,round(salary/12,2) as 'monthly salary' from employees;

first_name	last_name	monthly salary
Rahul	Sharma	4583.33
Priya	Mehta	5166.67
Amit	Patel	4000.00
Neha	Verma	4333.33
Rohan	Chauhan	5416.67
Sneha	Parihar	4833.33
Karan	Joshi	3916.67
Pooja	Bansal	4166.67
Vikas	Singh	6000.00
Anjali	Kapoor	4083.33
Arjun	Shetty	5666.67
Kavita	Shukla	4500.00
Nitin	Kher	3833.33
Riya	Chopra	5083.33
Mohit	Bhardwaj	4250.00
Simran	Kaur	4666.67
Varun	Bedi	4916.67
Nisha	Desai	4416.67
Saurabh	Jain	6250.00
Divya	Malhotra	4750.00

20 rows in set (0.00 sec)